



**A 15 ps Two-Dimensional Vernier
Time-to-Digital Converter in 180 nm BCD**

Design report
Digital IC Design II (EE4615)

5-bit thermometer output, trimmed resolution 15.0–15.85 ps, validated in all five process corners.
Schematic and Spectre simulation only; every cell hand-designed in TSMC 180 nm BCD.

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Abstract

This report presents the design and verification of a 5-bit two-dimensional (2-D) Vernier time-to-digital converter (TDC) in TSMC 180 nm BCD. Two delay lines with $\tau_1 = 90$ ps and $\tau_2 = 75$ ps span a 10×6 comparison grid whose 31 thermometer levels each map to exactly one SR-latch arbiter, giving a resolution $t_0 = \tau_1 - \tau_2 = 15$ ps, below the 20 ps specification and below the technology gate delay. With one fixed trim setting the converter passes all sign-off checks in all five process corners (LSB 13.1–18.2 ps, worst DNL 0.90 LSB, no missing codes); per-corner MOSCAP trimming re-centers the LSB to 15.0–15.85 ps, a six times tighter corner spread. The core consumes 11.7 pJ per conversion at a figure of merit of 0.50 pJ per step within a total transistor width of 1305.6 μm , and the arbiter dead zone is below 1 fs. A 1-D Vernier row built from the same cells confirms the resolution and area advantage of the 2-D architecture.

1 Introduction

1.1 Background and motivation

A time-to-digital converter (TDC) quantizes the interval between two signal edges. In an all-digital PLL it replaces the analog phase detector and charge pump, which made TDCs standard in digital RF synthesis [4, 7]. The basic flash TDC, a delay line sampled along its taps, has a resolution floor of one gate delay. In 180 nm CMOS that floor is several tens of picoseconds under realistic loading (our loaded stage bottoms out at 97 ps with conventional fan-out-of-four, FO=4, sizing), above the 20 ps this project requires, so a sub-gate-delay technique is mandatory.

The Vernier principle reaches below the gate delay by racing two delay lines: the START edge propagates through stages of delay τ_1 while the STOP edge chases it through faster stages of delay τ_2 . The effective resolution is the difference $t_0 = \tau_1 - \tau_2$, which can be made arbitrarily small in principle [3]. The classical one-dimensional implementation pays for this with hardware and latency: one stage pair and one arbiter per code, so range and resolution fight each other. Ring and interpolation variants recover range at the cost of control or matching complexity [5, 6]. The two-dimensional (2-D) Vernier of Liscidini, Vercesi and Castello [1, 2] instead compares *every* START tap against *every* STOP tap, so a grid of $N_X + N_Y$ stages covers $\mathcal{O}(N_X N_Y)$ distinct intervals. We chose this architecture because it decouples range from line length at a low cell count.

1.2 Objectives and boundary conditions

The EE4615 brief sets hard requirements and a design-quality trade-off (Table 1). Every gate must be hand-designed in TSMC 180 nm BCD; standard cells are not allowed. Verification is by Spectre simulation of the schematic, no layout. The course testbench fixes the interface: a 40 ps RESET pulse, START and STOP inputs driven through $3\times$ minimum inverters, and 31 thermometer outputs loaded by the TDC_out cell [8]. Robustness must be demonstrated in the five process corners TT, SS, FF, SF and FS [9].

Table 1: Project requirements and quality metrics.

Requirement	Value
Output	5 bit, thermometer code $Q_{1..31}$
Resolution (LSB)	better than 20 ps
Range	full 32-code linear range
Process / supply	TSMC 180 nm BCD, 1.8 V
Corners	TT, SS, FF, SF, FS, all must pass
Standard cells	none; all cells hand-designed
Verification	schematic + Spectre only
Quality metric	Definition
Linearity	DNL, INL per corner
Energy	average energy per conversion
Area	total transistor width ΣW
Figure of merit	energy per conversion step

1.3 Contributions and report structure

The delivered converter reaches a trimmed resolution of 15.0–15.85 ps in all five corners with no missing codes. Three results carry the design: FO=4 sizing cannot reach the required stage delays in this technology, delay cells must be sized against the true in-grid load, and a measured 1-D versus 2-D comparison built from identical cells isolates the architectural benefit from the circuit design.

Section 2 develops the architecture and the operating point. Section 3 documents every cell with its transistor sizes. Section 4 describes the design iterations and the verification setup. Section 5 reports the corner, trimming, energy, metastability and comparison results, and Section 6 concludes.

2 Architecture

2.1 From one to two dimensions

In a 1-D Vernier TDC the STOP edge gains $t_0 = \tau_1 - \tau_2$ on the START edge at every stage, and an arbiter at each stage records whether STOP has caught up. Resolving the 31 thermometer levels therefore takes 31 stages in each line and 31 arbiters, and the conversion latency grows with the full 31 τ_1 . The 2-D Vernier [2] generalizes the race: with START applied to a line of N_X stages of delay τ_1 and STOP to a line of N_Y stages of delay τ_2 , an arbiter at grid position (i, j) compares the edge leaving START stage i with the edge leaving STOP stage j and fires when

$$\Delta t > i \tau_1 - j \tau_2. \quad (1)$$

Each cell therefore measures its own threshold, and $N_X + N_Y$ stages yield up to $N_X N_Y$ distinct thresholds. Range no longer dictates line length.

2.2 Bijective level mapping at $k = 6$

We restrict the design space to the family $\tau_1 = k t_0$, $\tau_2 = (k - 1) t_0$. In this family the threshold of cell (i, j) equals $(ik - j(k - 1)) t_0$, an exact integer multiple of the LSB, and the mapping from thermometer level m to grid cell is bijective [2]:

$$y_m = ((m - 1) \bmod k) + 1, \quad x_m = \frac{m + (k - 1) y_m}{k}, \quad m = 1, \dots, 31. \quad (2)$$

Level m is exactly one SR latch, wired straight to output Q_m . No OR tree or encoder sits between arbiter and output, so combining-logic skew cannot corrupt the code. For $k = 6$ the mapping spans $x_m \leq 10$ and $y_m \leq 6$: a 10×6 logical grid, 10 START stages and 6 STOP stages for all 31 levels.

The operating point follows from two constraints. The LSB must clear the 20 ps specification with margin for corner spread, which sets $t_0 = 15$ ps. The stage delays must stay inside the window that the delay cell can actually produce under its in-grid load (Section 4), which selects $k = 6$: $\tau_1 = 90$ ps and $\tau_2 = 75$ ps. Smaller k would require τ_2 below the achievable minimum; larger k would lengthen both lines and add area without improving resolution.

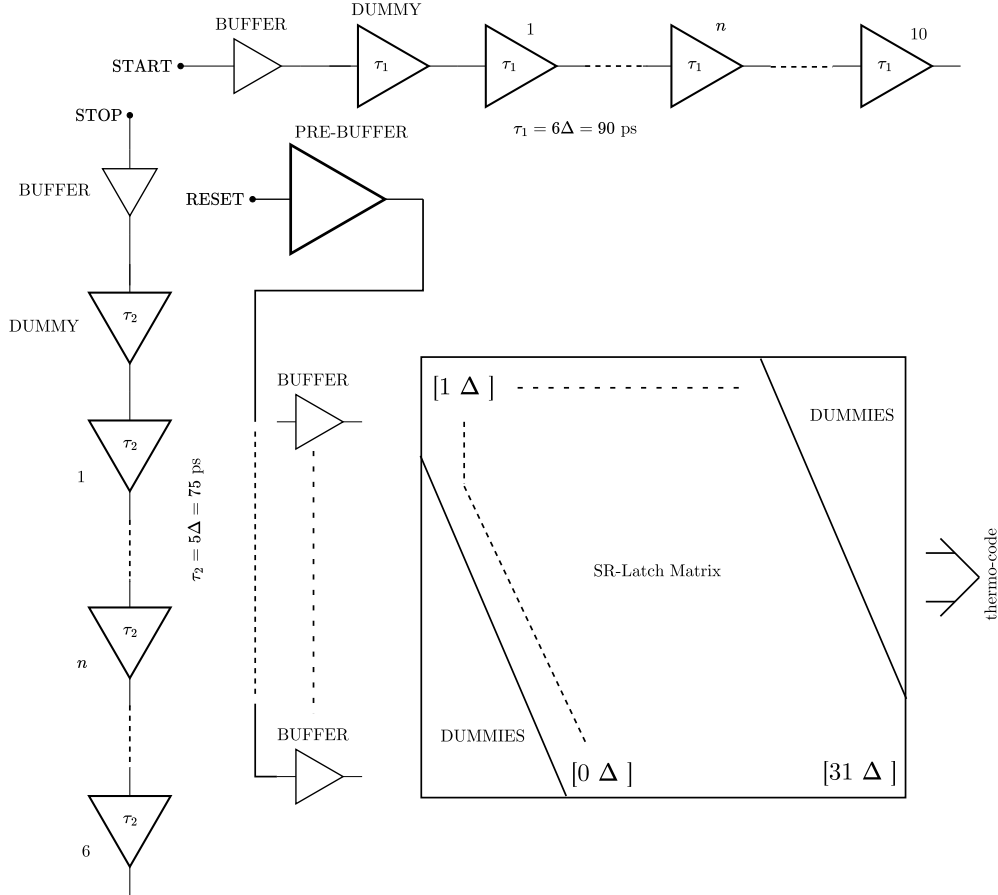


Figure 1: Top level of the converter. The START line (10 stages of τ_1 , top) and the STOP line (6 stages of τ_2 , left) feed the SR-latch matrix. The 31 active latches sit on the bijective map of Eq. (2); dummy latches fill the unused grid cells so the load is uniform along each line. Buffer trees distribute START, STOP and the 40 ps RESET. The thermometer code is read directly from the latch outputs.

2.3 Fan-out equalization and reset

Equation (1) only holds if every stage of a line contributes the same delay, and stage delay depends on load. With only the 31 mapped latches in place, the taps would drive anywhere from one to ten latch inputs, which would modulate the stage delays along each line and corrupt the thresholds. We therefore fill the unused cells of the 10×6 array with 29 dummy latches, electrically identical to the active ones but with unused outputs: every START tap then drives exactly six latch inputs and every STOP tap exactly ten, so the load is uniform along each line. The same argument adds one extra delay cell at the head of each line (11 and 7 cells in total), so that the first logical stage sees the same driver as all others.

Each latch carries an asynchronous clear input. The 40 ps RESET pulse from the testbench,

widened through its own buffer tree, clears all 60 latches before each conversion so that every conversion starts from $Q_m = 0$.

3 Circuit design

All cells were designed at transistor level in the `vernier2d` library using the 2 V devices of the `tsmc18` library (`nmos2v/pmos2v`, Spectre models `nch/pch`) at the minimum channel length $L = 180$ nm. The unit inverter used throughout is $W_p/W_n = 440$ nm/220 nm; larger drivers are built as parallel fingers (multiples) of this unit. Table 2 summarizes the cell inventory; the per-cell width budget is given in Appendix B.

Table 2: Cell inventory of the TDC core with device sizing. All lengths 180 nm except the MOSCAPs (250 nm).

Cell	Count	Contents and sizing
<code>delay_tau1</code>	11	FO=1 inverter pair (10× input, 10× driver, unit 440/220 nm); MOSCAPs for trimming: three TG-switched trim banks of 2/20/21 fingers and one fixed TG-MOSCAP (7/18 fingers) connection to ensure $\tau_1 = \tau_2 + 15$ ps.
<code>delay_tau2</code>	7	identical active devices as <code>delay_tau1</code> ; lighter trim banks of 2/14/18 fingers.
<code>srlatch</code>	60	13 transistors: two 2-input NANDs (all devices 440/180 nm), asynchronous-reset NMOS (2.0/0.18 μ m), two unit-inverter output buffers.
<code>nand</code> <code>Inv_10x/20x/40x/80x</code>	(in latch) trees	two series NMOS, two parallel PMOS, all 440/180 nm. input and reset distribution buffers, parallel-finger multiples of the unit inverter.

3.1 Delay stage

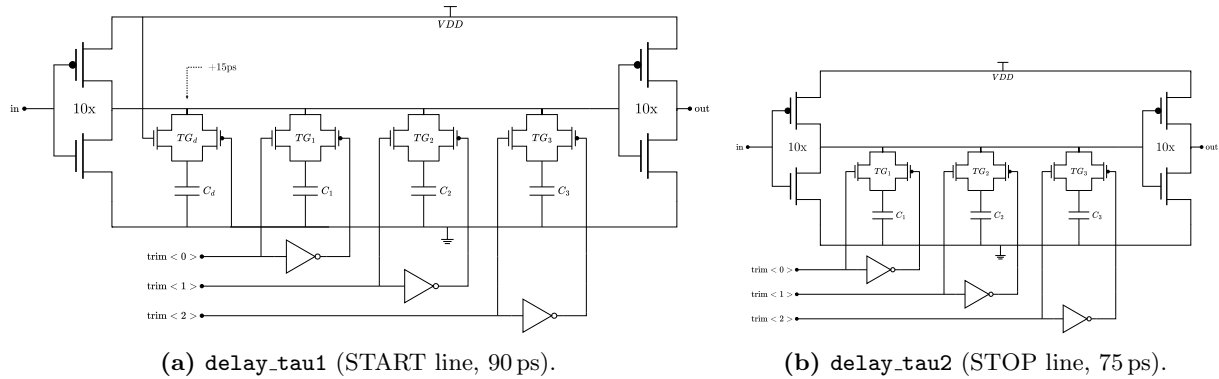


Figure 2: Delay stages. An FO=1 inverter pair drives the next stage; the internal node carries one fixed MOSCAP behind an always-on access transistor plus three transmission-gate-switched MOSCAP banks controlled by `trim<0:2>`. The two cells differ only in bank sizes.

The delay stage is a two-inverter buffer whose internal node is loaded capacitively (Fig. 2). Sizing it is a compromise. Larger devices average out mismatch and make the stage less sensitive to parasitics and load variation, which protects linearity and corner stability; the same width, however, costs area and power and adds self-loading, which slows the stage and can push it outside the usable delay window. The chosen size is therefore the smallest one that still meets the 75 ps and 90 ps targets under the true grid load of Section 4.1.

The internal fan-out follows the same logic. Conventional FO=4 sizing minimizes the delay

of a logic path, but here delay is the product, not the cost: with $FO=4$ the intrinsic delay of the internal node pins the minimum achievable stage delay at 97 ps in this technology, beyond the required τ_2 . Equal input and driver sizing ($10\times$ and $10\times$ unit fingers, $FO=1$) reduces the internal-node capacitance and opens a usable window of 74–147 ps under the true grid load.

Trimming is capacitive, with identical actives. Both lines use the same active devices; only the capacitance on the internal node differs. The fixed MOSCAP ($18\times$ fingers, $L = 250$ nm) sets the nominal delay, and three banks (2/20/21 fingers on the τ_1 cell, 2/14/18 on the τ_2 cell) are switched in or out through transmission gates by the static `trim<0:2>` inputs. Because both lines share the same $10\times$ $FO=1$ actives, τ_1 and τ_2 track each other over process and temperature, which is what keeps their difference, the LSB, stable. At the trimmed typical-corner point the lines measure $\tau_1 = 90.5$ ps and $\tau_2 = 75.4$ ps.

3.2 Arbiter

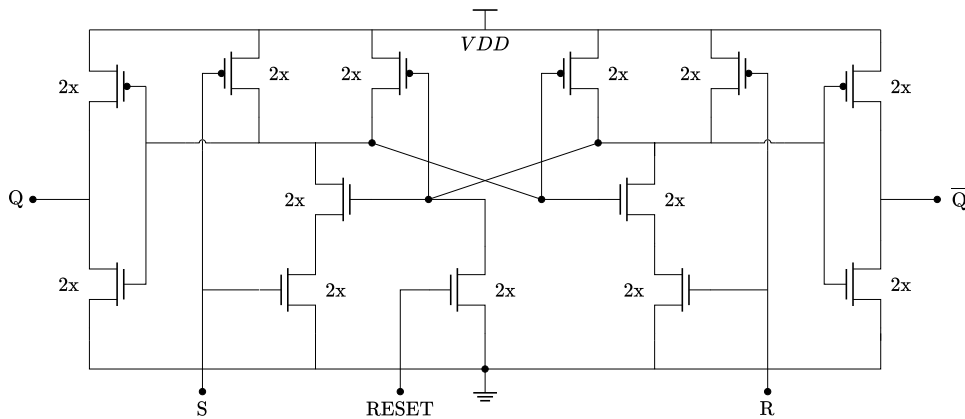


Figure 3: SR-latch arbiter (`vernier2d/srlatch`). Two cross-coupled NANDs (440/180 nm throughout), an asynchronous-reset NMOS on the internal node, and two unit-inverter output buffers. S and R are active high; the first rising edge wins.

The arbiter decides, at each grid node, which edge arrived first. We use a cross-coupled NAND SR latch with buffered outputs (Fig. 3), 13 transistors in total. Both NAND outputs idle high; the first rising input flips the latch and positive feedback locks the decision. Three properties motivated the choice over a D flip-flop. The cell is small, which matters at 60 instances. The S and R paths are electrically symmetric with equal NMOS and PMOS widths, which is what keeps the arbitration offset at the femtosecond level measured in Section 5.6. And reset costs a single NMOS ($2.0/0.18\ \mu\text{m}$) that pulls one internal node during the 40 ps RESET pulse, defining $Q = 0$ before every conversion.

3.3 Core assembly

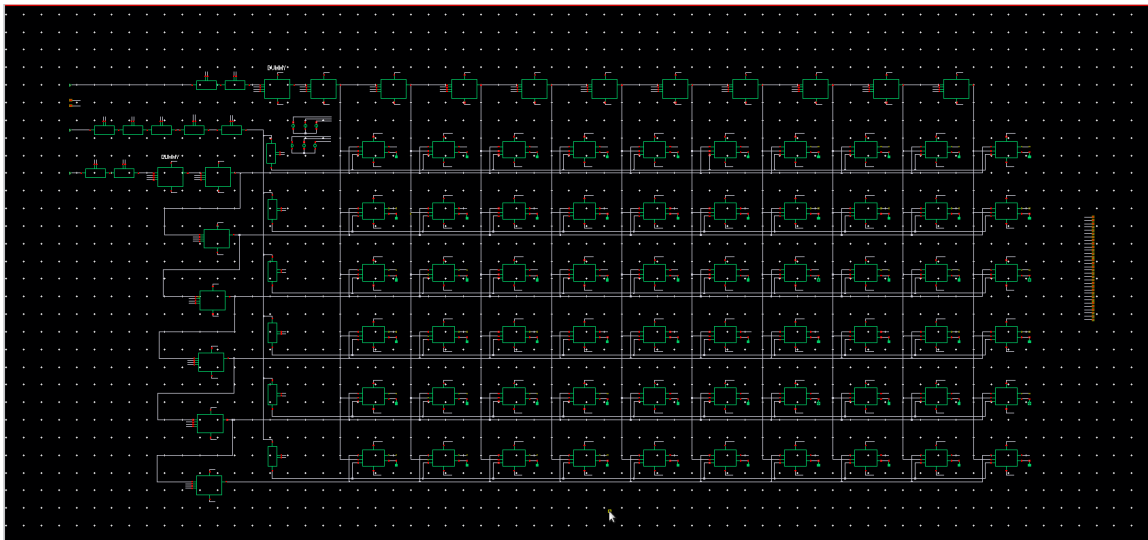


Figure 4: `vernier2d/tdc_core` as entered in Cadence Virtuoso: START line across the top, STOP line down the left, and the 10×6 padded latch matrix. The core contains 1120 active devices.

The core (Fig. 4) instantiates 11 `delay_tau1` and 7 `delay_tau2` stages, 60 latches and the input buffer trees, and exposes the course-mandated pins (RESET, START, STOP, V_{DD} , GND, $Q_{1..31}$). The six trim lines enter as static gate biases, standing in for configuration bits. The total transistor width is $\Sigma W = 1305.6 \mu\text{m}$: $1078.3 \mu\text{m}$ of active devices (1120 transistors) and $227.3 \mu\text{m}$ of MOSCAP (65 devices). The width budget is dominated by the `delay_tau1` drivers ($451 \mu\text{m}$) and the latch NANDs ($211 \mu\text{m}$); the full breakdown is in Appendix B.

4 Design methodology and verification setup

4.1 Sizing against the real load

The decisive step of the design was recognizing what a delay stage actually drives inside the grid. A τ_2 tap feeds the latch inputs of an entire row, a τ_1 tap those of an entire column, and both feed the next delay stage, whose input capacitance scales with the stage sizing itself. After fan-out equalization the equivalent load is about twenty unit-inverter inputs per tap.

Stage sizing derived from a standalone sweep with an idealized load predicted $t_0 = 16.6 \text{ ps}$; the same cells assembled into a row with raw tap loading measured 28.6 ps (Section 5.7), nearly a factor of two lost to loading. All final sizing was therefore done against a replica load of ten latch inputs plus one downstream stage, swept in the slowest corner (SS) so that the slow line stays reachable everywhere. The same analysis moved the architecture itself: an earlier $k = 4$ operating point ($\tau_1/\tau_2 = 60/45 \text{ ps}$) required stage delays below the achievable window and was abandoned in favor of $k = 6$.

4.2 Per-corner trimming

Process corners shift both line delays together, but not perfectly, so t_0 breathes with the corner. The MOSCAP banks of Section 3.1 absorb this. All 64 on/off combinations of the six bank gates were simulated per corner, measuring τ_1 and τ_2 directly on the tap nets; the configuration with t_0 closest to 15 ps was selected and validated end-to-end with a full staircase sweep (Appendix A). Trimming changes gate voltages only, so the transistor inventory and ΣW are identical in every corner.

4.3 Testbench and measurement method

All system-level results come from the unmodified course testbench TB.TDC [8]: stimulus generation in TDC_in, the DUT wrapper, and TDC_out loading every thermometer output with two $2\times$ inverters and 1 fF. The DUT receives its supply through a dedicated rail with a series ammeter, separate from the stimulus rail, so the measured energy, $E = V_{DD} \int |i| dt$ over each 5 ns conversion window, contains only the converter.

OCEAN automates the characterization. Per corner, the input delay is swept in 1.5 ps steps (one tenth of the LSB) across the full range, and the scripts extract the LSB (mean step width), DNL, INL, offset, energy, ENOB and the sign-off checks: no missing codes, monotonicity, and an intact thermometer code at every point. The course script reports INL as the half-range value $(INL_{\max} - INL_{\min})/2$, while the plots show the endpoint-referenced per-code curve; we quote the course convention for sign-off and state both where they differ.

5 Results

All results are Spectre simulations of the schematic at 27 °C and $V_{DD} = 1.8$ V through the course testbench, in the five corners TT, SS, FF, SF and FS.

5.1 Transfer and resolution across corners

Figure 5 shows the measured transfer with one fixed trim configuration (the trimmed TT setting) in all five corners; Table 3 lists the extracted metrics. Every corner produces a monotonic 31-step staircase with an intact thermometer code, and every corner passes the 20 ps requirement, with the slowest corner (SS, 18.2 ps) closest to the limit. The skewed corners SF and FS barely differ from TT because opposite N and P shifts largely cancel in an inverter pair. The input-referred offset stays between -7 and -24 ps, constant per corner: a code offset, not a linearity error.

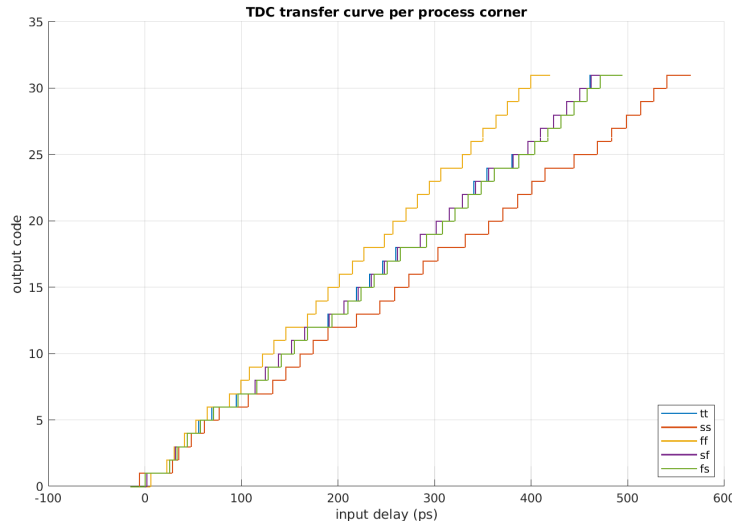


Figure 5: TDC transfer per corner, fixed trim configuration. All corners are monotonic and reach code 31; the slope spreads with corner speed.

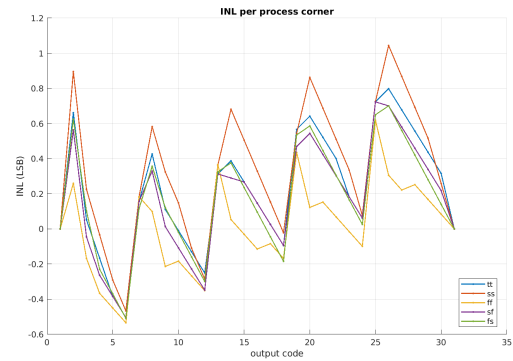
Table 3: Corner results with one fixed trim configuration. INL is the course half-range convention; energy and FoM from the supply-ammeter measurement. Every corner passes all sign-off checks (no missing codes, monotonic, LSB below 20 ps).

Corner	LSB (ps)	DNL _{max} (LSB)	INL (LSB)	Offset (ps)	E/conv (pJ)	ENOB (bit)	FoM (pJ/step)
TT	15.35	0.66	0.65	−15.4	11.67	4.54	0.50
SS	18.20	0.90	0.76	−24.2	10.76	4.39	0.51
FF	13.10	0.72	0.58	−7.1	12.84	4.58	0.54
SF	15.35	0.66	0.61	−13.9	11.96	4.57	0.50
FS	15.70	0.72	0.61	−15.7	11.38	4.56	0.48

5.2 Linearity



(a) DNL per corner.



(b) INL per corner (endpoint convention).

Figure 6: Linearity with fixed trim. The period-6 sawtooth marks the column-wrap boundaries of the $k = 6$ mapping and repeats identically in every corner.

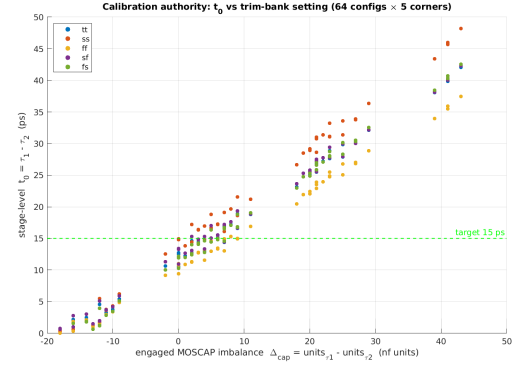
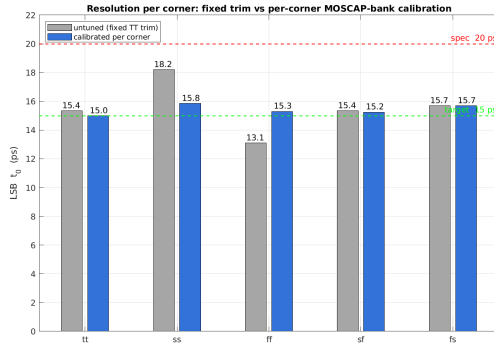
Worst-case DNL is 0.90 LSB (SS), well clear of the -1 LSB missing-code limit; no corner loses a code (Fig. 6). Switching from the half-range convention of Table 3 to the per-code endpoint curve of Fig. 6b, the worst endpoint INL is 1.04 LSB (SS); all other corners stay at or below 0.80. The dominant DNL feature is a sawtooth with period 6, peaking at 0.6–0.7 LSB at codes 7, 13, 19 and 25: the column-wrap boundaries of the mapping in Eq. (2). The pattern is identical in every corner, so it is architectural rather than mismatch-driven, and trimming cannot remove it (Section 5.3). A targeted capacitive correction at the wrap codes is listed as future work.

5.3 Per-corner trimming

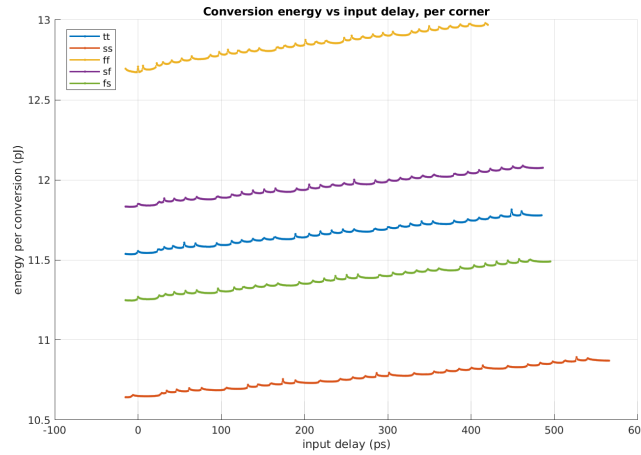
Applying the per-corner trim codes of Appendix A re-centers the LSB to 15.0–15.85 ps: the corner spread tightens from 5.10 ps to 0.85 ps, six times, and the worst-case margin to the specification grows from 1.8 to 4.15 ps (Fig. 7a). DNL and INL barely move (worst trimmed INL 0.74 LSB), consistent with the architectural origin of the wrap pattern. All sign-off checks still pass in every corner.

5.4 Energy, power and figure of merit

Energy per conversion sits in a flat 10.6–13.0 pJ band across input delay and corner (Fig. 8), ordered by corner speed (SS lowest, FF highest). With the 5 ns conversion window of the testbench this corresponds to an average power of 2.15–2.57 mW. Table 4 reports both figure-of-merit conventions: the specification’s P/t_0 and the course script’s $E/2^{\text{ENOB}}$. The dynamic range, $31 \times \text{LSB}$, spans 406–564 ps depending on the corner, with a quantization-limited single-shot precision $\text{LSB}/\sqrt{12}$ of 3.8–5.3 ps.



(a) LSB per corner, fixed trim vs per-corner trim.

(b) Trim authority: stage-level t_0 vs engaged capacitance imbalance, all 64 configurations in 5 corners.**Figure 7:** MOSCAP-bank trimming. Every corner re-centers to the 15 ps target; the banks command roughly 0–48 ps, far more authority than the corners require.**Figure 8:** Energy per conversion vs input delay, per corner. The band is nearly flat; energy tracks corner speed.**Table 4:** Power and figure of merit per corner (fixed trim). P_{avg} assumes the 5 ns conversion window (200 MS/s).

Corner	P_{avg} (mW)	P/t_0 (mW/ps)	$E/2^{ENOB}$ (pJ/step)
TT (nominal)	2.33	0.152	0.50
SS	2.15	0.118	0.51
FF	2.57	0.196	0.54
SF	2.39	0.156	0.50
FS	2.28	0.145	0.48

5.5 Area

The transistor-width total of the core is $\Sigma W = 1305.6 \mu\text{m}$: $1078.3 \mu\text{m}$ of active width over 1120 devices plus $227.3 \mu\text{m}$ of MOSCAP over 65 devices, extracted from the Spectre netlist. Including the testbench interface cells would add $93.7 \mu\text{m}$ (TDC_in $11.9 \mu\text{m}$, TDC_out $81.8 \mu\text{m}$). The per-cell breakdown is given in Appendix B.

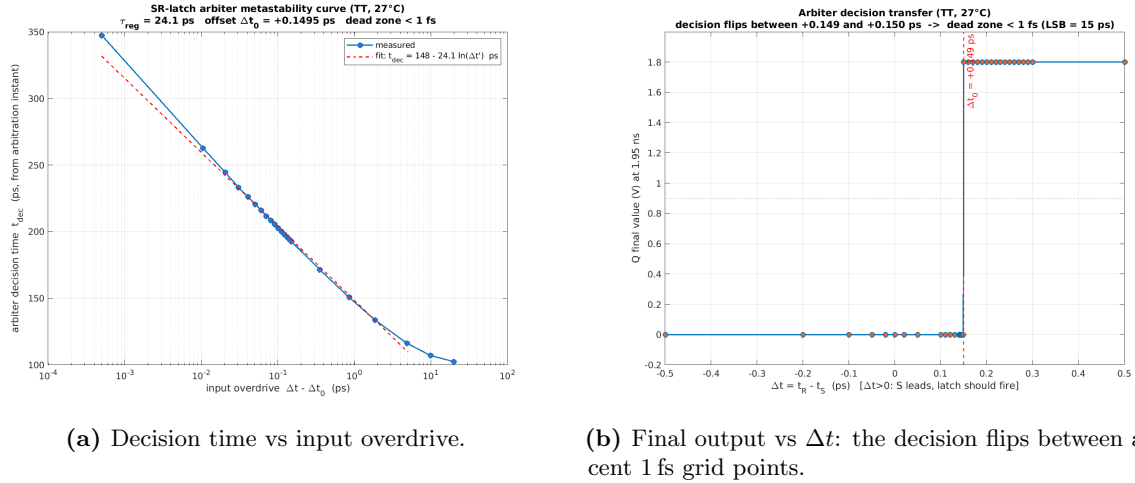
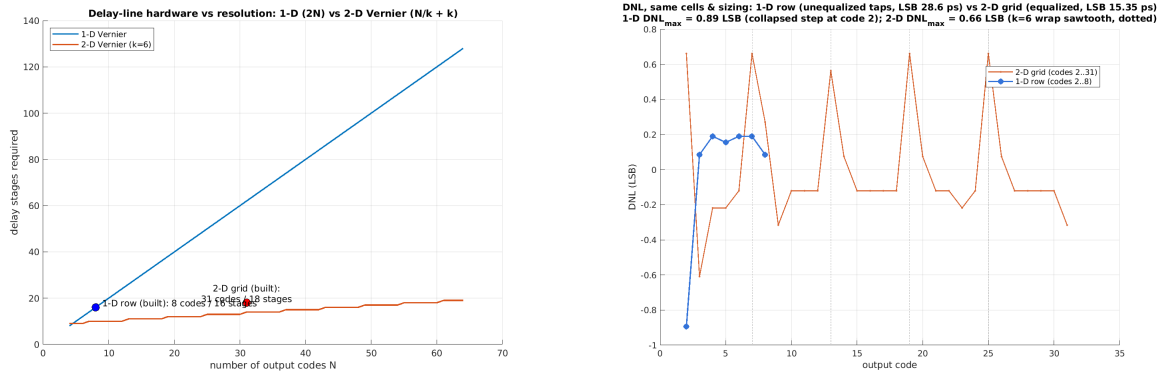


Figure 9: SR-latch arbiter characterization (TT, 27°C, 5 fF loads), 49 simulations in three nested sweeps.

5.6 Arbiter dead zone and metastability

The smallest resolvable interval of the converter is set by the arbiter, so we characterized it down to 0.5 fs of input overdrive (Fig. 9). The arbitration offset is $\Delta t_0 = +149.5 \pm 0.5$ fs, about 1% of one LSB. The decision time follows the classic regeneration law $t_{dec} = t_{cq} + \tau_{reg} \ln(1/\Delta t')$ over three and a half decades with $\tau_{reg} = 24.1$ ps and a large-overdrive clk-to-Q of 102 ps; the worst observed decision time is 347.5 ps at 0.5 fs overdrive, still far inside the conversion window. No ambiguous final value was observed anywhere on the 1 fs grid, bounding the dead zone below 1 fs, four orders of magnitude under the LSB. The bound is grid-limited and noiseless (a Monte-Carlo analysis would widen it), which still leaves orders of magnitude of margin.

5.7 Measured comparison with a 1-D Vernier



(a) Delay stages required vs code count, with both built designs marked.

(b) DNL, 1-D row vs 2-D core, same cells.

Figure 10: Architecture comparison from identical cells (TT).

To separate the architecture from the circuit design we assembled an 8-code 1-D Vernier row (`vernier2d/vernier1d_row`) from the same delay and latch cells, with trim banks hardwired on and, deliberately, no fan-out equalization. Table 5 compares the measurements. The row resolves 28.6 ps where the stage-level prediction was 16.6 ps, confirming the loading analysis of Section 4.1, and the 2-D core is about twice as area-efficient per code at comparable energy. Scaled to the full 31 codes, the row would need $3.4\times$ the delay stages, $1.9\times$ the width and 3.3 ns of full-scale latency against 0.9 ns for the grid.

Table 5: Measured 1-D row vs 2-D core, identical cells (TT, 27 °C).

	1-D row (built)	2-D core (built)
Codes	8	31
LSB	28.6 ps	15.35 ps
DNL _{max}	0.89 LSB	0.66 LSB
Delay stages	16	18
ΣW per code	81.4 μm	42.1 μm
Energy per code	0.33 pJ	0.38 pJ
Full-scale latency (31 codes)	3.3 ns	0.9 ns

6 Conclusion

We built and verified a 5-bit 2-D Vernier TDC in TSMC 180 nm BCD that meets every project requirement with margin: with one fixed trim setting it resolves 13.1–18.2 ps across all five process corners with no missing codes, and per-corner trimming re-centers the LSB to 15.0–15.85 ps, a six times tighter spread, without changing a single device.

Three conclusions follow from the evidence. First, delay-cell sizing for Vernier grids must be done against the in-grid load; the measured 1-D row shows the LSB doubling when tap loading is ignored. Second, filling the grid with dummy latches is what makes the thresholds usable; it costs 29 inactive cells and buys corner-independent linearity. Third, the bijective $k = 6$ mapping removes all combining logic, and the only systematic linearity signature left, the period-6 wrap sawtooth, is architectural and identical in every corner.

Recommendations. A DLL-style background loop could set the trim codes automatically, and a voltage-to-time front end would extend the TDC to voltage quantization. The most direct improvement, however, is a targeted capacitive correction at the four column-wrap codes, which follows immediately from the measured DNL pattern. Layout and extracted re-simulation are the natural next step toward silicon; the margins reported here leave realistic room for parasitic degradation.

References

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A Per-corner trim configurations

Trimming sets the six trim-bank gates (`trim<0:2>` of each line) to 0 V or 1.8 V. Table 6 lists the winning configuration per corner from the exhaustive 64-point probe, together with the stage delays it produces. The FS optimum coincides with the default setting.

Table 6: Trim codes per corner and the stage delays they produce. In the code $b_2b_1b_0$, bank i is engaged when $b_i = 1$ (gate at 1.8 V); bank weights ($b_0/b_1/b_2$) are 2/20/21 fingers on the τ_1 cell and 2/14/18 on the τ_2 cell.

Corner	τ_1 code	τ_2 code	τ_1 (ps)	τ_2 (ps)	t_0 (ps)
TT	010	010	88.5	73.4	15.04
SS	001	001	93.0	78.0	14.91
FF	111	111	90.5	75.6	14.97
SF	100	011	89.8	74.7	15.07
FS	011	011	92.6	77.6	15.01

The stage-level probe slightly underestimates the staircase LSB (SS: 14.91 ps probe vs 15.85 ps measured) because grid loading beyond the first stage adds a corner-dependent few percent. This is why every configuration was re-validated with a full staircase sweep.

B Transistor-width budget

Table 7 breaks the netlist-extracted ΣW of `tdc_core` down per cell, summed over all instantiations.

Table 7: ΣW breakdown of the core (Spectre netlist, TT trim setting). MOSCAPs identified by drain, source and bulk tied together. The grand total is quoted as 1305.6 μm in the text.

Group	ΣW (μm)
delay_tau1 (11 inst.)	450.89
delay_tau2 (7 inst.)	207.34
nand (latch gates)	211.20
srlatch (reset devices)	120.00
Inv (latch output buffers)	114.84
Inv_20x	92.40
Inv_80x	52.80
Inv_40x	26.40
Inv_10x	19.80
Inv_5x	9.90
Active total (1120 devices)	1078.32
MOSCAP trim banks (65 devices)	227.25
Grand total (1185 devices)	1305.57

C Simulation inventory

All results are reproducible from the OCEAN scripts submitted with the design.

- **Corner campaign** (`tdc_corners_all.ocn`): five corners, 1.5 ps delay step, 1670 transients; produces the transition tables behind Figs. 5–6 and Table 3.

- **Trim probe** (`tune_probe.ocn`): 64 trim configurations $\times$ 5 corners, 320 transients; produces Fig. 7b and Table 6.
- **Trimmed campaign** (`tdc_tuned_all.ocn`): repeats the corner campaign with per-corner trim codes; produces Fig. 7a.
- **Metastability** (`meta_sweep_{coarse,fine,ultra}.ocn`): three nested sweeps, 49 transients, overdrive resolved to 0.5 fs; produces Fig. 9.
- **1-D comparison** (`v1d_staircase_v2.ocn`, `v1d_energy2.ocn`): 321-run staircase and energy measurement of the 1-D row; produces Fig. 10 and Table 5.